

## Assignments Terraform Overview

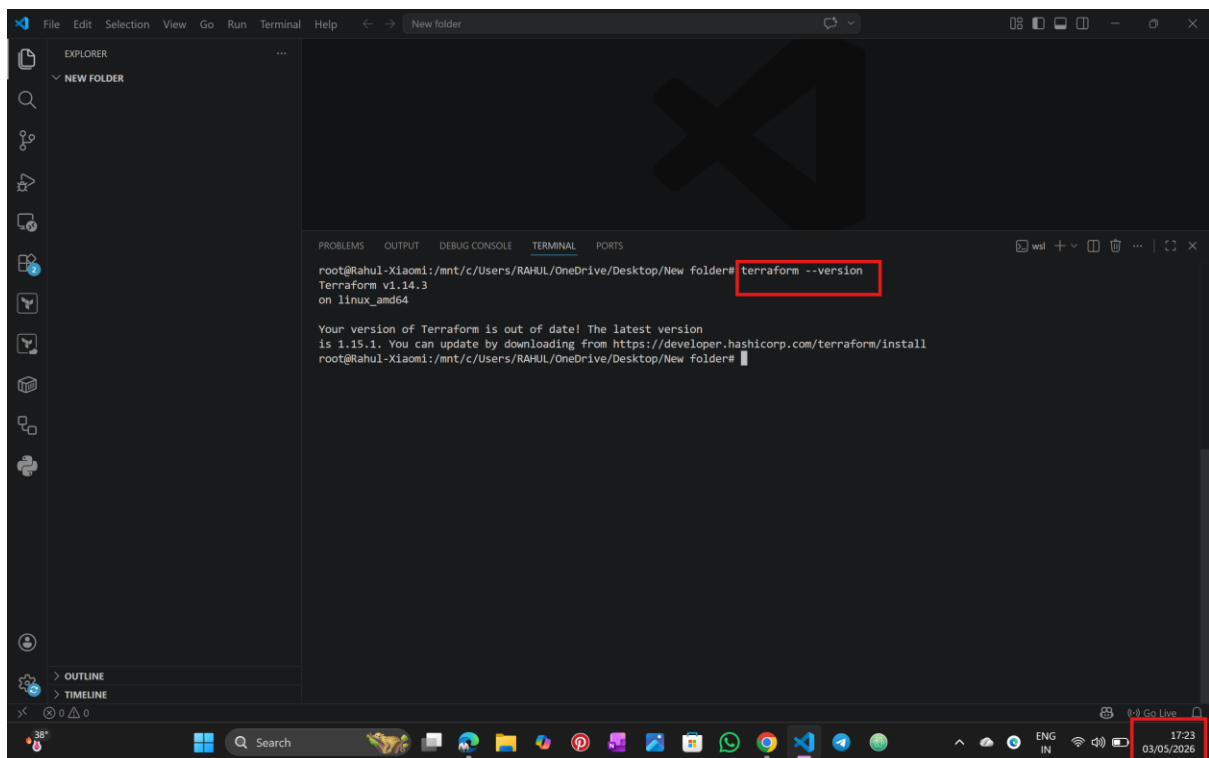
Submitted to - Vikul Sir

Date of Submission – 28/APR/2026

Submitted by – Rahul Nalathati

L1 - Provision AWS EC2 Instance along with VPC, Subnet, Internet Gateway, Route-Table, RouteTable Association, Security Group

Step :1 Install terraform in your desktop and access using VSC

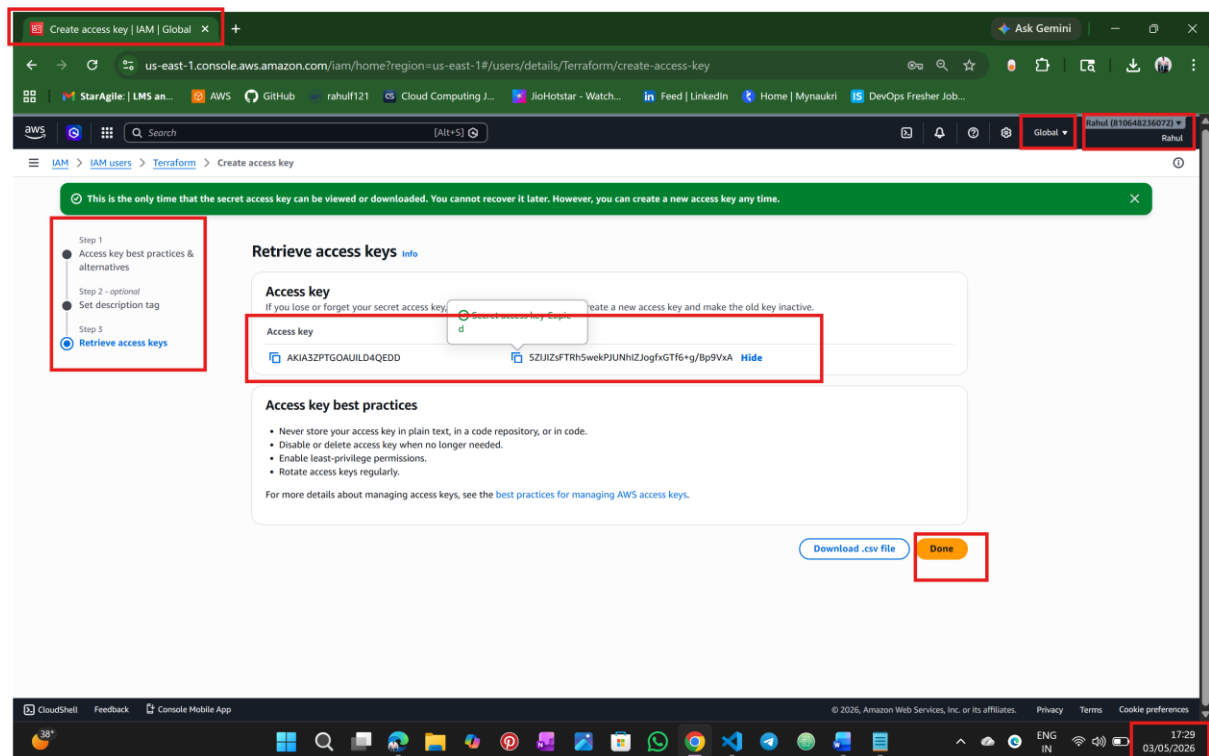
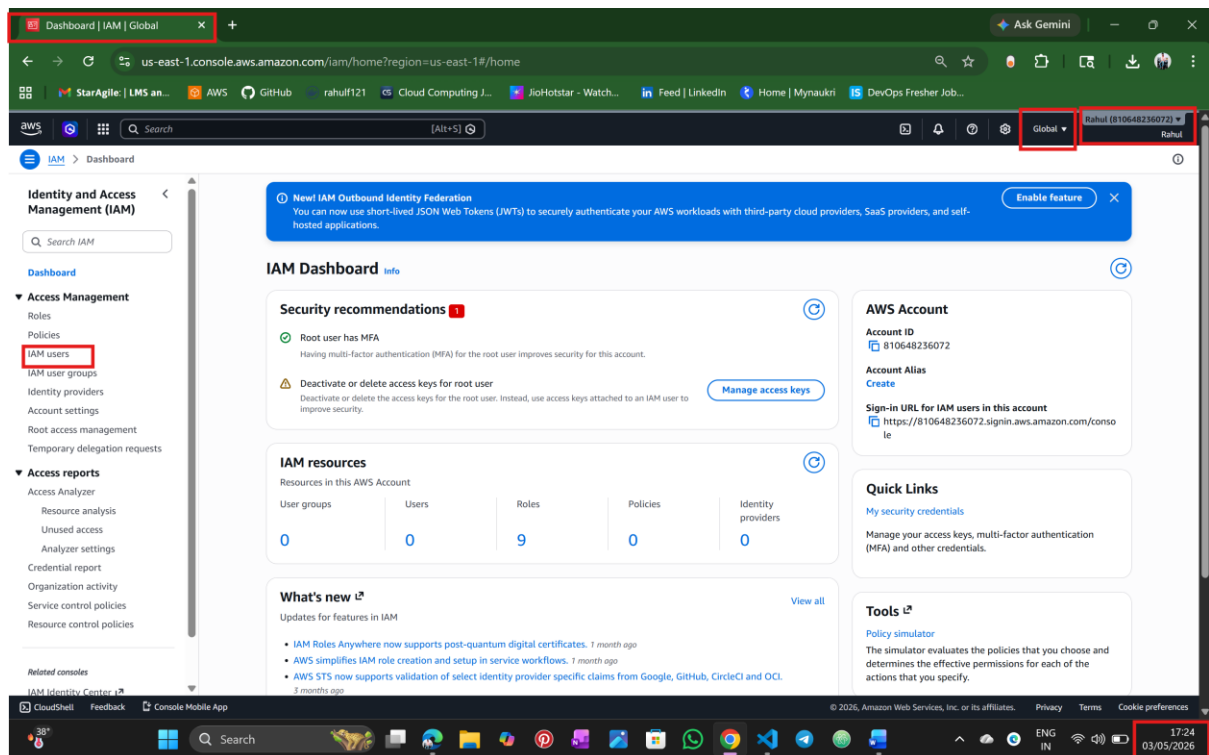


The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal is running a command prompt session. The command `terraform --version` has been entered and executed. The output shows the current version of Terraform is v1.14.3 on linux\_amd64. A message indicates that the current version is out of date and the latest version is 1.15.1, with a link to the download page. The terminal window is titled 'New folder' and the command prompt is `root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder#`. The system tray at the bottom right shows the time as 17:23 on 03/05/2026.

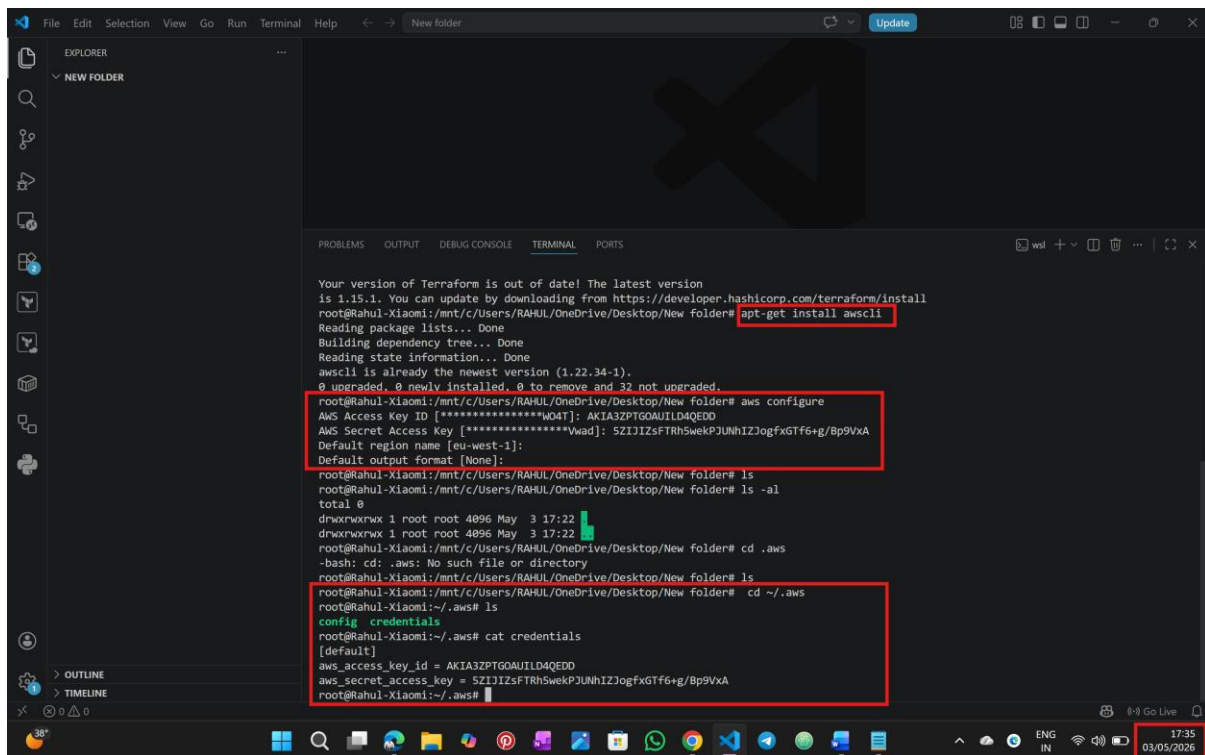
```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder# terraform --version
Terraform v1.14.3
on linux_amd64

Your version of Terraform is out of date! The latest version
is 1.15.1. You can update by downloading from https://developer.hashicorp.com/terraform/install
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder#
```

## Step :2 create a IAM user and access key and Secret Key



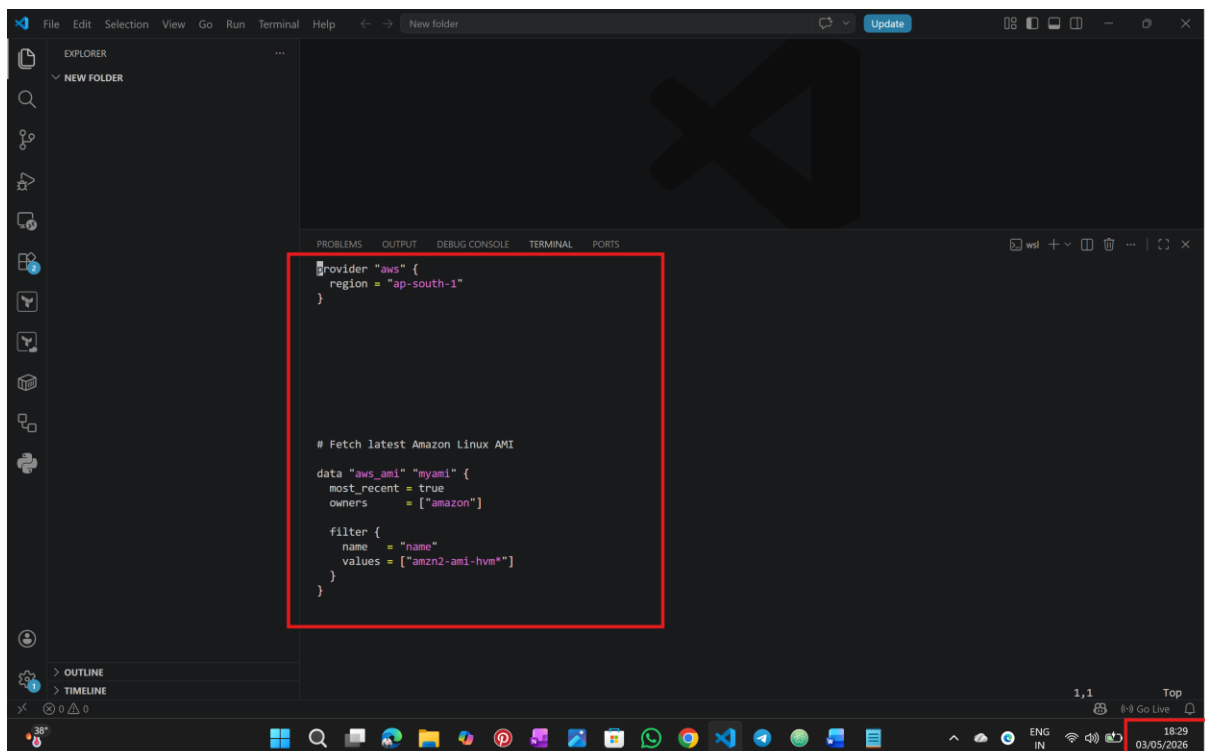
### Step: 3 configure the AWS credentials



```
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder# apt-get install awscli
Your version of Terraform is out of date! The latest version
is 1.15.1. You can update by downloading from https://developer.hashicorp.com/terraform/install
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
awscli is already the newest version (1.22.34-1).
0 upgraded, 0 newly installed, 0 to remove and 32 not upgraded.
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder# aws configure
AWS Access Key ID [*****@QAT]: AKIA3ZPTG0AU1LD4QEDD
AWS Secret Access Key [*****Vwad]: 5ZIJIZsFTRhSwekPJUNhIZJogfxGf6+g/Bp9VxA
Default region name [eu-west-1]:
Default output format [None]:
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder# ls
total 0
drwxrwxrwx 1 root root 4096 May  3 17:22
drwxrwxrwx 1 root root 4096 May  3 17:22
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder# cd .aws
-bash: cd: .aws: No such file or directory
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder# ls
root@Rahul-Xiaomi:/mnt/c/Users/RAHUL/OneDrive/Desktop/New folder# cd ~/.aws
root@Rahul-Xiaomi:~/aws# ls
config  credentials
root@Rahul-Xiaomi:~/aws# cat credentials
[default]
aws_access_key_id = AKIA3ZPTG0AU1LD4QEDD
aws_secret_access_key = 5ZIJIZsFTRhSwekPJUNhIZJogfxGf6+g/Bp9VxA
root@Rahul-Xiaomi:~/aws#
```

### Step :4 Write the Terraform code to create infrastructure in AWS

→provider AWS



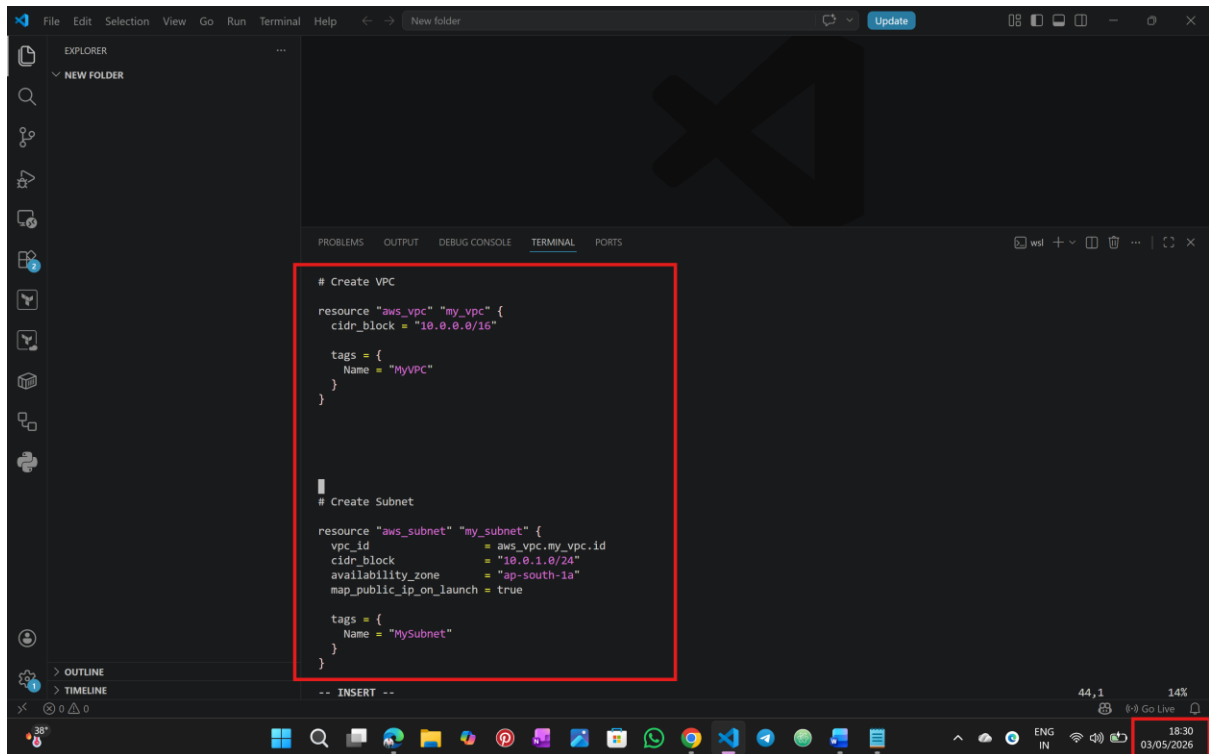
```
provider "aws" {
  region = "ap-south-1"
}

# Fetch latest Amazon Linux AMI
data "aws_ami" "myami" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["amzn2-ami-hvm*"]
  }
}
```

→VPC

→subnets



The screenshot shows the Visual Studio Code editor with a Terraform configuration file. The code is written in HCL and defines two resources: an AWS VPC and an AWS Subnet. The VPC resource is named 'my\_vpc' and has a cidr\_block of '10.0.0.0/16'. The Subnet resource is named 'my\_subnet' and is associated with the VPC. The Subnet has a cidr\_block of '10.0.1.0/24' and is located in the 'ap-south-1a' availability zone. The code is highlighted with a red box.

```
# Create VPC
resource "aws_vpc" "my_vpc" {
  cidr_block = "10.0.0.0/16"

  tags = {
    Name = "MyVPC"
  }
}

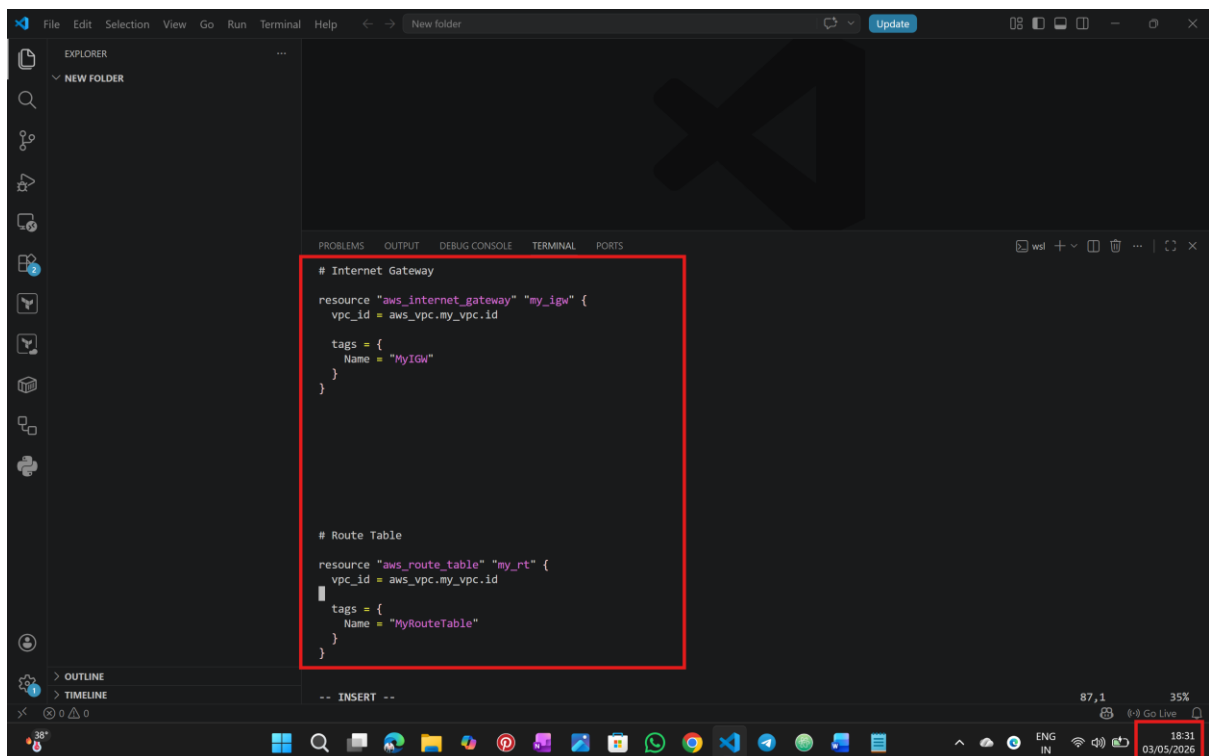
# Create Subnet
resource "aws_subnet" "my_subnet" {
  vpc_id      = aws_vpc.my_vpc.id
  cidr_block  = "10.0.1.0/24"
  availability_zone = "ap-south-1a"
  map_public_ip_on_launch = true

  tags = {
    Name = "MySubnet"
  }
}
```

The status bar at the bottom right shows the file size as 44,1 and the encoding as 14%. The system tray at the bottom right shows the time as 18:30 and the date as 03/05/2026.

→internet gateway

→route table



The screenshot shows the Visual Studio Code editor with a Terraform configuration file. The code is written in HCL and defines two resources: an AWS Internet Gateway and an AWS Route Table. The Internet Gateway resource is named 'my\_igw' and is associated with the VPC. The Route Table resource is named 'my\_rt' and is associated with the VPC. The code is highlighted with a red box.

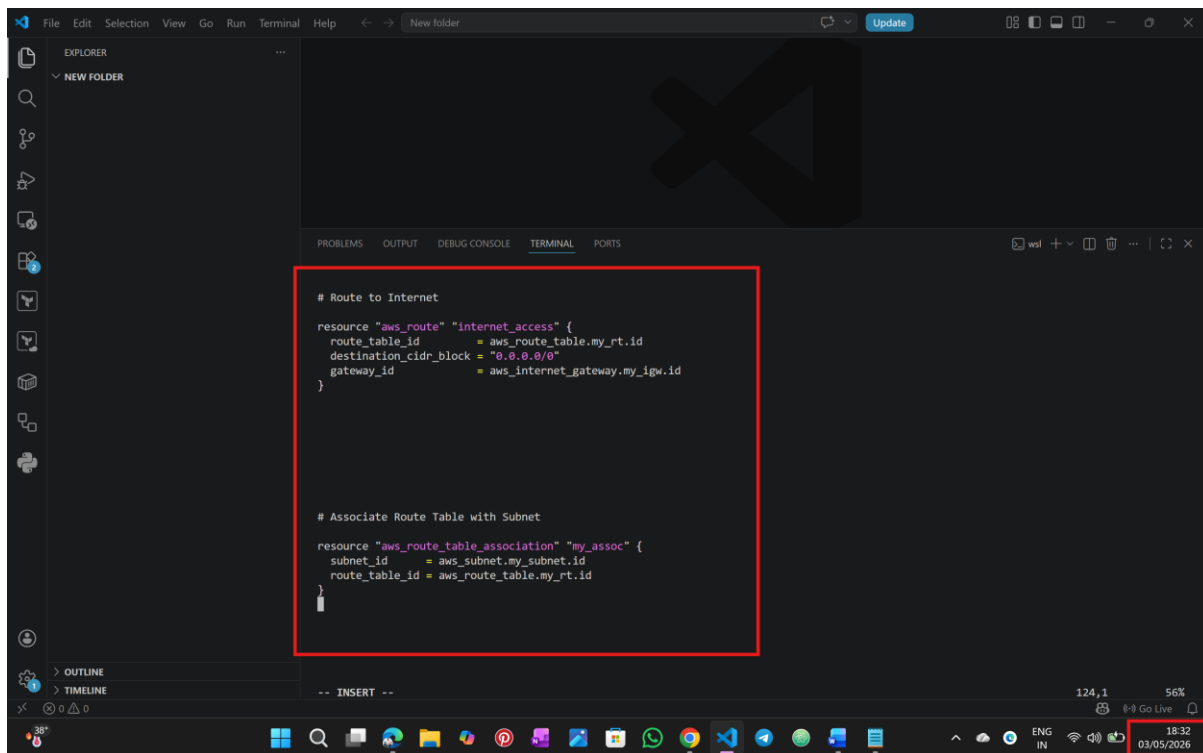
```
# Internet Gateway
resource "aws_internet_gateway" "my_igw" {
  vpc_id = aws_vpc.my_vpc.id

  tags = {
    Name = "MyIGW"
  }
}

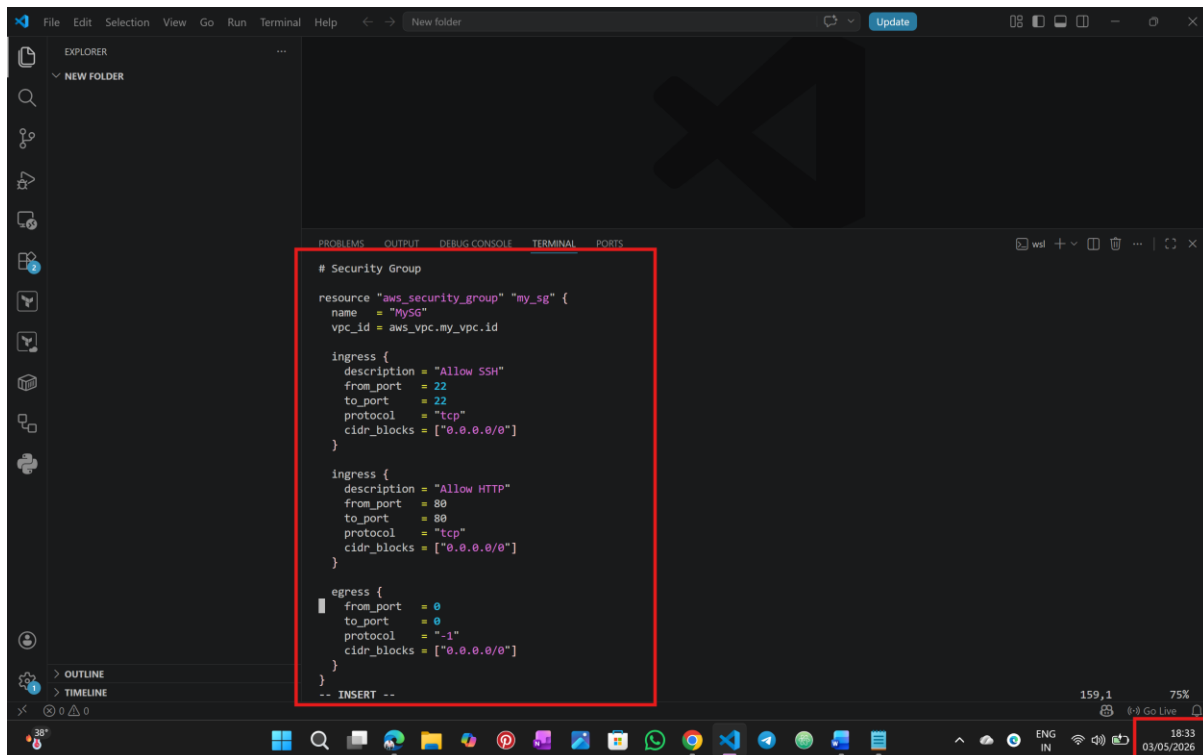
# Route Table
resource "aws_route_table" "my_rt" {
  vpc_id = aws_vpc.my_vpc.id

  tags = {
    Name = "MyRouteTable"
  }
}
```

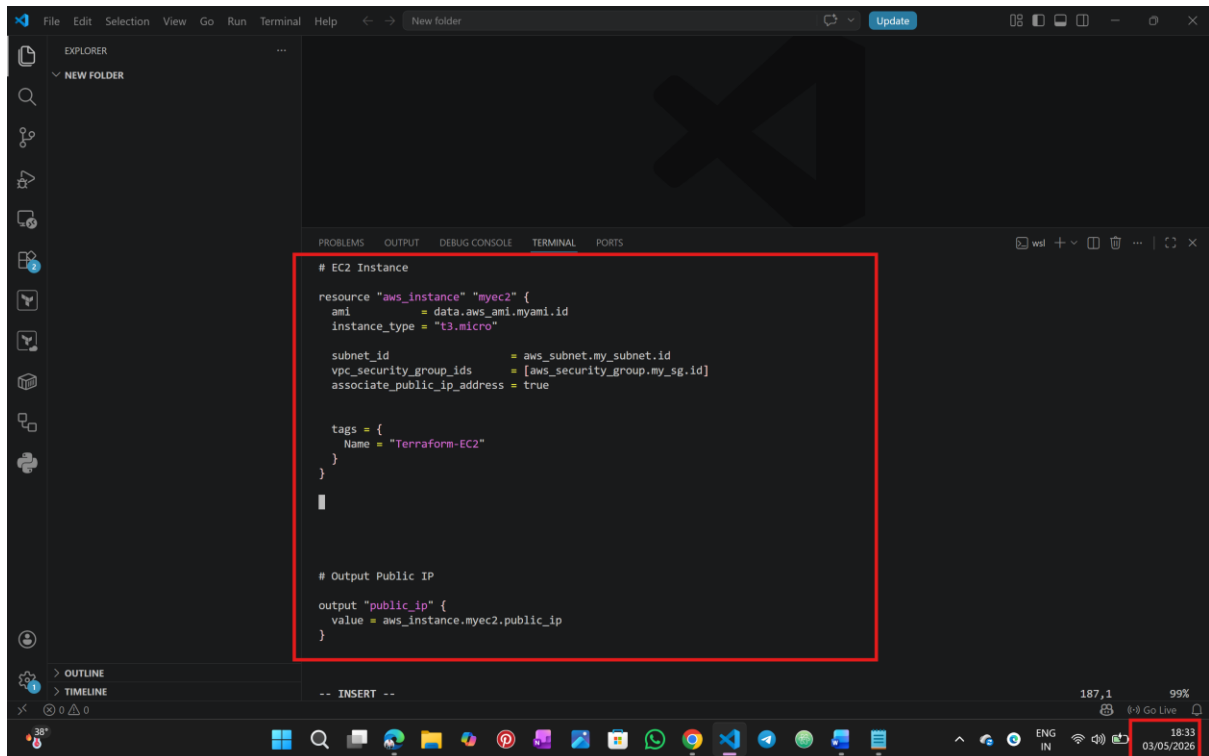
The status bar at the bottom right shows the file size as 87,1 and the encoding as 35%. The system tray at the bottom right shows the time as 18:31 and the date as 03/05/2026.



## →Security Group



## →Ec2 instances



The screenshot shows a Visual Studio Code editor window with a Terraform configuration file. The configuration is for an EC2 instance named 'myec2'. It uses the 'aws\_instance' resource with the following attributes: 'ami' is 'data.aws\_ami.myami.id', 'instance\_type' is 't3.micro', 'subnet\_id' is 'aws\_subnet.my\_subnet.id', 'vpc\_security\_group\_ids' is '[aws\_security\_group.my\_sg.id]', and 'associate\_public\_ip\_address' is 'true'. The tags are set to 'Name = "Terraform-EC2"'. An output block 'public\_ip' is defined with the value 'aws\_instance.myec2.public\_ip'.

```
# EC2 Instance

resource "aws_instance" "myec2" {
  ami           = data.aws_ami.myami.id
  instance_type = "t3.micro"

  subnet_id      = aws_subnet.my_subnet.id
  vpc_security_group_ids = [aws_security_group.my_sg.id]
  associate_public_ip_address = true

  tags = {
    Name = "Terraform-EC2"
  }
}

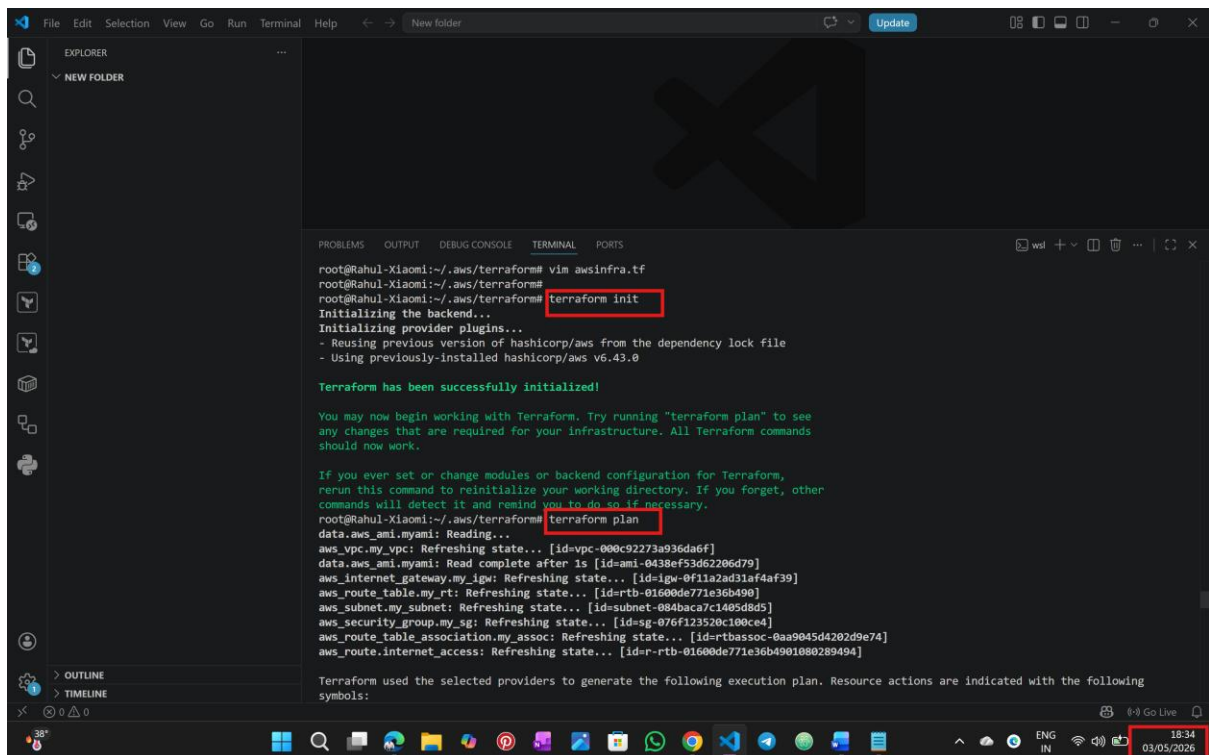
# Output Public IP

output "public_ip" {
  value = aws_instance.myec2.public_ip
}
```

## →terraform init

## →terraform plan

## →terraform apply



The screenshot shows the terminal output of the 'terraform init' and 'terraform plan' commands. The 'terraform init' command successfully initializes the Terraform environment, reusing the previous version of the dependency lock file and the previously installed hashicorp/aws v6.43.0 provider. The 'terraform plan' command shows the state of the resources, including 'data.aws\_ami.myami', 'aws\_vpc.my\_vpc', 'aws\_internet\_gateway.my\_igw', 'aws\_route\_table.my\_rt', 'aws\_subnet.my\_subnet', 'aws\_security\_group.my\_sg', 'aws\_route\_table\_association.my\_assoc', and 'aws\_route.internet\_access'. The plan shows that the resources are being read or refreshed.

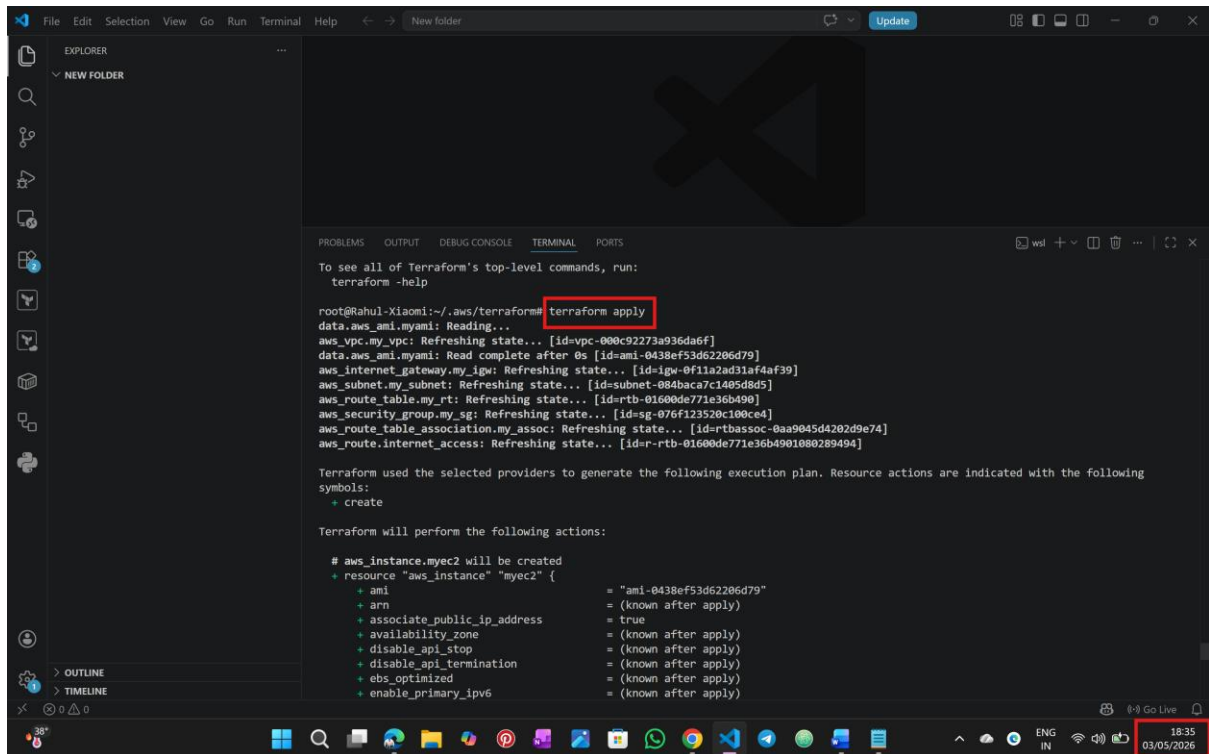
```
root@Rahul-Xiaomi:~/aws/terraform# vim awsinfra.tf
root@Rahul-Xiaomi:~/aws/terraform# terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.43.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
root@Rahul-Xiaomi:~/aws/terraform# terraform plan
data.aws_ami.myami: Reading...
aws_vpc.my_vpc: Refreshing state... [id=vpc-000c92273a936da6f]
data.aws_ami.myami: Read complete after 1s [id=ami-0438ef33d62206d79]
aws_internet_gateway.my_igw: Refreshing state... [id=igw-0f11a2ad31af4f39]
aws_route_table.my_rt: Refreshing state... [id=rtb-01600de771e36b490]
aws_subnet.my_subnet: Refreshing state... [id=subnet-084baca7c1405d8d5]
aws_security_group.my_sg: Refreshing state... [id=sg-076f123520c100ce4]
aws_route_table_association.my_assoc: Refreshing state... [id=rtbassoc-0aa9045d4202d9e74]
aws_route.internet_access: Refreshing state... [id=r-rtb-01600de771e36b4901080289494]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
```



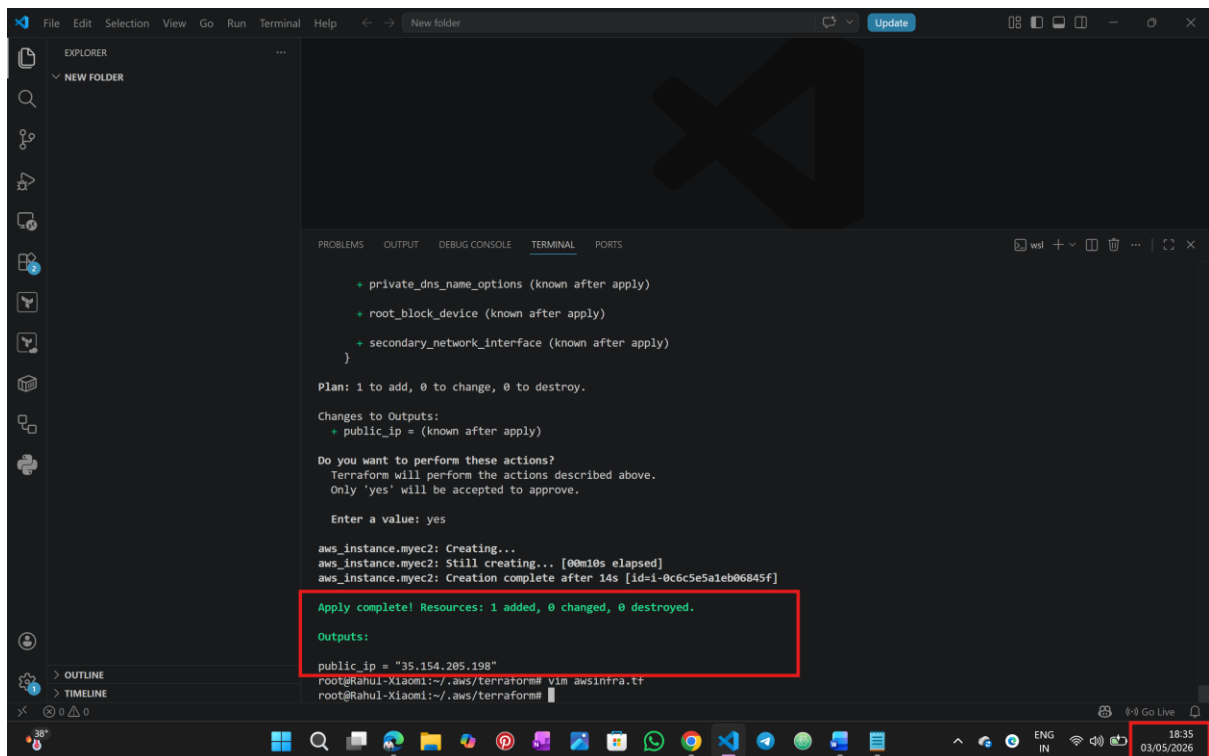
The screenshot shows a VS Code window with a terminal. The terminal is running the command `terraform apply`. The output shows the state of various AWS resources being refreshed, including `data.aws_ami.myami`, `aws_vpc.my_vpc`, `aws_internet_gateway.my_igw`, `aws_subnet.my_subnet`, `aws_route_table.my_rt`, `aws_security_group.my_sg`, `aws_route_table_association.my_assoc`, and `aws_route.internet_access`. The output also indicates that Terraform will perform the following actions: `# aws_instance.myec2 will be created` and `+ resource "aws_instance" "myec2" {`. The terminal output is as follows:

```
root@Rahul-Xiaomi:~/aws/terraform# terraform apply
data.aws_ami.myami: Reading...
aws_vpc.my_vpc: Refreshing state... [id=vpc-000c92273a936da6f]
data.aws_ami.myami: Read complete after 0s [id=ami-0438ef53d62206d79]
aws_internet_gateway.my_igw: Refreshing state... [id=igw-0f11a2ad31afaf39]
aws_subnet.my_subnet: Refreshing state... [id=subnet-004bca7c1486d8d5]
aws_route_table.my_rt: Refreshing state... [id=rtb-01600de771e36b490]
aws_security_group.my_sg: Refreshing state... [id=sg-076f123520c190ce4]
aws_route_table_association.my_assoc: Refreshing state... [id=rthassoc-0aa9045d4202d9e74]
aws_route.internet_access: Refreshing state... [id=r-rtb-01600de771e36b4901080289494]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.myec2 will be created
+ resource "aws_instance" "myec2" {
  + ami                    = "ami-0438ef53d62206d79"
  + arn                    = (known after apply)
  + associate_public_ip_address = true
  + availability_zone       = (known after apply)
  + disable_api_stop        = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized           = (known after apply)
  + enable_primary_ipv6     = (known after apply)
```



The screenshot shows a VS Code window with a terminal. The terminal is running the command `terraform apply`. The output shows the state of various AWS resources being refreshed, including `data.aws_ami.myami`, `aws_vpc.my_vpc`, `aws_internet_gateway.my_igw`, `aws_subnet.my_subnet`, `aws_route_table.my_rt`, `aws_security_group.my_sg`, `aws_route_table_association.my_assoc`, and `aws_route.internet_access`. The output also indicates that Terraform will perform the following actions: `# aws_instance.myec2 will be created` and `+ resource "aws_instance" "myec2" {`. The terminal output is as follows:

```
root@Rahul-Xiaomi:~/aws/terraform# terraform apply
data.aws_ami.myami: Reading...
aws_vpc.my_vpc: Refreshing state... [id=vpc-000c92273a936da6f]
data.aws_ami.myami: Read complete after 0s [id=ami-0438ef53d62206d79]
aws_internet_gateway.my_igw: Refreshing state... [id=igw-0f11a2ad31afaf39]
aws_subnet.my_subnet: Refreshing state... [id=subnet-004bca7c1486d8d5]
aws_route_table.my_rt: Refreshing state... [id=rtb-01600de771e36b490]
aws_security_group.my_sg: Refreshing state... [id=sg-076f123520c190ce4]
aws_route_table_association.my_assoc: Refreshing state... [id=rthassoc-0aa9045d4202d9e74]
aws_route.internet_access: Refreshing state... [id=r-rtb-01600de771e36b4901080289494]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.myec2 will be created
+ resource "aws_instance" "myec2" {
  + ami                    = "ami-0438ef53d62206d79"
  + arn                    = (known after apply)
  + associate_public_ip_address = true
  + availability_zone       = (known after apply)
  + disable_api_stop        = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized           = (known after apply)
  + enable_primary_ipv6     = (known after apply)
  + private_dns_name_options = (known after apply)
  + root_block_device        = (known after apply)
  + secondary_network_interface = (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ public_ip = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.myec2: Creating...
aws_instance.myec2: Still creating... [00m10s elapsed]
aws_instance.myec2: Creation complete after 14s [id=i-0c6c5e5a1eb06845f]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:
public_ip = "35.154.205.198"
root@Rahul-Xiaomi:~/aws/terraform# vim awsinfra.tf
root@Rahul-Xiaomi:~/aws/terraform#
```

## Step :5 Result, check the all resources

The screenshot displays the AWS Management Console for the 'ap-south-1' region. The 'Instances' page is active, showing a list of EC2 instances. A table with the following columns is visible: Name, Instance ID, Instance type, Status check, Alarm status, Availability zone, and Public IPv4 DNS. One instance, 'Terraform-EC2' with ID 'i-0c6c5e5a1eb06845f', is highlighted in blue and has a red box around it. Below the table, the details for this instance are shown, including its state as 'Running' and its public IPv4 address '35.154.205.198'. The top navigation bar shows the user 'Rahul' and the region 'Asia Pacific (Mumbai)'. The bottom status bar shows the time '18:42' and date '03/05/2026'.

The screenshot displays the AWS Management Console for the 'ap-south-1' region, specifically the 'VPCs' page. A table lists VPCs with columns: Name, VPC ID, State, Encryption controls, Encryption control, Block Public IP, IPv4 CIDR, IPv6 CIDR, and DHCP option set. Two VPCs are listed: 'vpc-000c92273a936da6f' and 'MyVPC' (vpc-000c92273a936da6f). The 'MyVPC' entry is highlighted in blue and has a red box around it. Below the table, the details for 'MyVPC' are shown, including a diagram of its resources: 'Subnets (1)' (ap-south-1a, MySubnet), 'Route tables (2)' (MyRouteTable, rtb-0262f4427370e0078), and 'Network Connections (1)' (MyIGW). The top navigation bar shows the user 'Rahul' and the region 'Asia Pacific (Mumbai)'. The bottom status bar shows the time '18:44' and date '03/05/2026'.



Subnets (1/4) Info

Find subnets by attribute or tag

| Name     | Subnet ID                | State     | VPC                           | Block Public... | IPv4 CIDR      |
|----------|--------------------------|-----------|-------------------------------|-----------------|----------------|
| MySubnet | subnet-0601a5557f441ddb7 | Available | vpc-08e8b8dde8e06223a         | Off             | 172.31.32.0/20 |
|          | subnet-084baca7c1405d8d5 | Available | vpc-000c92273a936da6f   MyVPC | Off             | 10.0.1.0/24    |
|          | subnet-077bdad108f3791a  | Available | vpc-08e8b8dde8e06223a         | Off             | 172.31.16.0/20 |
|          | subnet-092f500e21169746c | Available | vpc-08e8b8dde8e06223a         | Off             | 172.31.0.0/20  |

subnet-084baca7c1405d8d5 / MySubnet

Details | Flow logs | Route table | Network ACL | CIDR reservations | Sharing | Tags

Details

Subnet ID  
subnet-084baca7c1405d8d5

Subnet ARN  
arn:aws:ec2:ap-south-1:810648236072:subnet/subnet-084baca7c1405d8d5

State  
Available

Block Public Access  
Off

IPv4 CIDR  
10.0.1.0/24

Available IPv4 addresses  
250

IPv6 CIDR  
-

VPC  
vpc-000c92273a936da6f | MyVPC

Route table  
-

18:47 03/05/2026

Route tables (1/3) Info

Find route tables by attribute or tag

| Name         | Route table ID        | Explicit subnet associ... | Edge associations | Main | VPC                           |
|--------------|-----------------------|---------------------------|-------------------|------|-------------------------------|
| MyRouteTable | rtb-0e95075619edce9e2 | -                         | -                 | Yes  | vpc-08e8b8dde8e06223a         |
|              | rtb-01600de771e36b490 | subnet-084baca7c1405d8d5  | -                 | No   | vpc-000c92273a936da6f   MyVPC |
|              | rtb-02d2f4d27370e0078 | -                         | -                 | Yes  | vpc-000c92273a936da6f   MyVPC |

rtb-01600de771e36b490 / MyRouteTable

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Details

Route table ID  
rtb-01600de771e36b490

Main  
No

Explicit subnet associations  
subnet-084baca7c1405d8d5 / MySubnet

VPC  
vpc-000c92273a936da6f | MyVPC

Owner ID  
810648236072

Edge associations  
-

18:48 03/05/2026

igw-vpc | ap-south-1 | Instances | EC2 | ap-south-1 | aws\_instance | Resources | hash | + | Ask Gemini

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#igws:

StarAgile | LMS an... | AWS | GitHub | rahul121 | Cloud Computing J... | JioHotstar - Watch... | Feed | LinkedIn | Home | Mynaukri | DevOps Fresher Job...

ecR | Ask Amazon Q

Asia Pacific (Mumbai) | Rahul (810648236072) | Rahul

VPC > Internet gateways

VPC dashboard < | AWS Global View | Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Internet gateways (1/2) info

Find internet gateways by attribute or tag

|                                     | Name  | Internet gateway ID   | State    | VPC ID                        | Owner        |
|-------------------------------------|-------|-----------------------|----------|-------------------------------|--------------|
| <input type="checkbox"/>            | -     | igw-004852948f2b74cd7 | Attached | vpc-08e8b8dde8e06223a         | 810648236072 |
| <input checked="" type="checkbox"/> | MyIGW | igw-0f11a2ad31af4af39 | Attached | vpc-000c92273a936da6f   MyVPC | 810648236072 |

igw-0f11a2ad31af4af39 / MyIGW

Details | Tags

Details

Internet gateway ID | sg-076f123520c100ce4 | State | Attached | VPC ID | vpc-000c92273a936da6f | MyVPC | Owner | 810648236072

CloudShell | Feedback | Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy | Terms | Cookie preferences

18:48 03/05/2026

VPC | ap-south-1 | Instances | EC2 | ap-south-1 | aws\_instance | Resources | hash | + | Ask Gemini

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SecurityGroups:

StarAgile | LMS an... | AWS | GitHub | rahul121 | Cloud Computing J... | JioHotstar - Watch... | Feed | LinkedIn | Home | Mynaukri | DevOps Fresher Job...

ecR | Ask Amazon Q

Asia Pacific (Mumbai) | Rahul (810648236072) | Rahul

VPC > Security Groups

VPC dashboard < | AWS Global View | Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Security Groups (1/3) info

Find security groups by attribute or tag

|                                     | Name | Security group ID    | Security group name | VPC ID                | Description                |
|-------------------------------------|------|----------------------|---------------------|-----------------------|----------------------------|
| <input checked="" type="checkbox"/> | -    | sg-076f123520c100ce4 | MySG                | vpc-000c92273a936da6f | Managed by Terraform       |
| <input type="checkbox"/>            | -    | sg-01af4a0bae0dc4d6c | default             | vpc-08e8b8dde8e06223a | default VPC security group |
| <input type="checkbox"/>            | -    | sg-01a0782d9daec4595 | default             | vpc-000c92273a936da6f | default VPC security group |

sg-076f123520c100ce4 - MySG

Details | Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Details

Security group name | MySG | Security group ID | sg-076f123520c100ce4 | Description | Managed by Terraform | VPC ID | vpc-000c92273a936da6f

Owner | 810648236072 | Inbound rules count | 2 Permission entries | Outbound rules count | 1 Permission entry

CloudShell | Feedback | Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy | Terms | Cookie preferences

18:50 03/05/2026

Rahul Nalathathi